Coordinate Geometry Maze

Name _____ Period ____

Start top-left. Give distances in simplest radical form. Rotations and dilations are centered at the origin; k is the scale factor. Go through the door with your answer.

START

Find the midpoint.

(2, 4) and (8, 10)

(5, 7)

Find the distance.

(1, 2) and (4, 6)

$\sqrt{7}$

Find the midpoint.

(−3, 5) and (7, −1)

(4, 4)

Reflect over the *y*-axis.

(−6, 2)

(3, 3)

5

7

(5, −3)

(10, −6)

4

(−6, −2)

Reflect over the *x*-axis.

(3, −2)

(3, 2)

Translate the point.

(−4, 1)

$(x, y) \to (x + 5, y - 3)$

(−9, −2)

Dilate from the origin.

(−2, 8), $k = \frac{1}{2}$

(−4, 16)

Find the slope.

perpendicular to

$y = -4x + 1$

(−3, −2)

(1, −2)

(1, 4)

(−9, 4)

$2\sqrt{5}$

(1, −4)

$-\frac{1}{4}$

Find the slope.

parallel to

$y = 3x - 4$

−3

Find the slope.

perpendicular to

$2x + 3y = 6$

$\frac{3}{2}$

Find the distance.

(−2, 3) and (4, −1)

$4\sqrt{13}$

Rotate about the origin.

(3, 4)

180°

3

(1, −5)

(−5, 2)

10

$2\sqrt{13}$

(5.5, −4.5)

(−4, 3)

Reflect over $y = x$.

(5, −1)

(−1, 5)

Rotate about the origin.

(2, 5)

90° counterclockwise

(5, −2)

Dilate from the origin.

(4, −6), $k = \frac{3}{2}$

(6, −9)

FINISH

You made it!

Coordinate Geometry Maze: Answer Grid

Name _____ Period ____

Write each answer in the order you found it, and show your work for that room.

1	Answer: _____	Work:
2	Answer: _____	Work:
3	Answer: _____	Work:
4	Answer: _____	Work:
5	Answer: _____	Work:
6	Answer: _____	Work:
7	Answer: _____	Work:
8	Answer: _____	Work:
9	Answer: _____	Work:
10	Answer: _____	Work: